
THE BLUEPRINT CHALLENGE

EV Battery Pack & Thermal Management System Design

12-Gate Engineering Pipeline Execution Directive

Executive Summary

The Blueprint must execute its complete 12-gate pipeline to design an EV battery pack and thermal management system optimized for maximum efficiency. This is not a research report — it is a full engineering execution that must produce a complete, actionable blueprint with evidence, alternatives, trade-offs, simulations, adversarial review, and honest uncertainty calibration.

The PatSnap result is an extremely good research report. It cites sources, identifies trade-offs, discusses patents, and acknowledges uncertainty. However, it does not produce a constraint graph, a complete trade-off engine, a benchmark score, adversarial review, confidence propagation, a manufacturing plan, a bill of materials, a repairability analysis, a cost model, a reachability analysis, simulations, or a complete decision tree.

The important question is not which system is smarter. The important question is: which system would an engineering organization trust more?

Challenge Input

The system shall receive the following input and must execute the complete pipeline to produce a full engineering blueprint. The system must not produce a generic summary. The system must behave as an engineering organization.

Design an EV battery pack and thermal management system optimized for maximum efficiency. Primary objective: maximize range per kWh. Secondary objectives: • minimize parasitic losses • minimize mass • maximize safety • maximize serviceability • maximize manufacturability • minimize total lifetime cost The system must not produce a generic summary. The system must behave as an engineering organization. The system shall execute the complete pipeline.

Gate 1 — Understanding

Produce:

- Objectives (primary + secondary, quantitative)
- Constraints (physical, economic, regulatory)
- Success criteria (measurable, falsifiable)
- Evaluation metrics (range/kWh, \$/kWh, kg/kWh, MTBF, yield)

Gate 2 — Evidence Collection

Minimum requirements:

- 20 patents (ranked C+)
- 20 papers (ranked D+)
- 10 production systems (ranked E+)
- 10 failures (ranked F+)
- 10 benchmarks (ranked E+)
- All classified and ranked per evidence hierarchy (A-I)

Gate 3 — Classification

Classify all evidence and components into:

- Energy storage
- Thermal management
- Structural engineering
- Manufacturing
- Control systems
- Safety engineering

Gate 4 — Component Decomposition

Produce complete decomposition of 15 components:

- Cells, Pack structure, Busbars, Contactors, Fuses
- Connectors, Cooling channels, Pumps, Valves, Sensors
- BMS, Insulation, Housing, Mounting systems, Fire suppression

Gate 5 — Alternatives Analysis

For every component, produce:

- Primary choice (with evidence and cost)

- Alternative A (with trade-off analysis)
- Alternative B (with trade-off analysis)
- Rejection criteria (why alternatives were rejected)

Gate 6 — Constraint Graph

Produce a directed acyclic graph (DAG) of constraint relationships:

- energy density → temperature → cooling power → range → cost
- Each edge has: relationship type, sensitivity score, description
- Identify critical paths through the constraint graph

Gate 7 — Trade-off Analysis

Evaluate across 7 dimensions:

- Cost, Mass, Reliability, Repairability, Manufacturability, Safety, Complexity
- For each design decision: show gains and sacrifices
- Net assessment: FAVORABLE / MARGINAL / UNFAVORABLE

Gate 8 — Benchmarking

Compare against 5 production systems:

- Tesla 4680 (structural pack, tabless cells)
- CATL Qilin (cell-to-pack, LFP+NCM hybrid)
- BYD Blade (LFP prismatic, structural)
- LG Energy Solution (NCM pouch, soft pack)
- Panasonic (NCM 18650/2170, cylindrical)
- 5 categories: best, cheapest, most reliable, most profitable, most scalable

Gate 9 — Failure Analysis

Study 7 failure modes:

- Thermal runaway (propagation, mitigation)
- Coolant leakage (detection, isolation)
- Seal failure (IP rating degradation)
- Swelling (cell expansion, structural stress)
- Corrosion (terminal, busbar, housing)
- Vibration (fatigue, connector loosening)
- Short circuits (internal, external, ground fault)
- Each: probability, severity, mitigation, evidence

Gate 10 — Simulation

Simulate 6 scenarios:

- Winter operation (-20°C, reduced capacity, heating demand)
- Summer operation (+45°C, thermal management stress)
- Rapid charging (350kW DC, thermal limits)
- Cell imbalance (capacity fade, resistance increase)
- Pump failure (coolant flow loss, thermal runaway risk)
- Sensor failure (BMS blind, conservative derating)
- Each: PASS / MARGINAL / FAIL with confidence and mitigation

Gate 11 — Adversarial Review

Four agents attempt to destroy the design:

- Electrical Engineer: challenges cell selection, busbar sizing, insulation
- Thermal Engineer: challenges cooling capacity, thermal runaway prevention
- Manufacturing Engineer: challenges yield, assembly time, tolerances
- Economist: challenges unit cost, break-even, margin, market assumptions
- Each reviewer must fail to find a fatal flaw

Gate 12 — Final Output

Produce complete blueprint:

- Architecture (system diagram, component layout)
- CAD specification (dimensions, materials, tolerances, joints)
- Bill of materials (all components, quantities, costs, suppliers)
- Manufacturing plan (assembly steps, yield, throughput)
- Maintenance plan (service intervals, spare parts, procedures)
- Supplier list (with alternatives and lead times)
- Cost model (capital, unit cost, break-even, ROI)
- Confidence score (with evidence, assumptions, penalties)
- Assumptions (with impact and falsifiers)
- Unknowns (with consequence and resolution path)
- Alternatives (for every major decision)
- Failure modes (with probability, severity, mitigation)

Scoring Rubric

The blueprint will be scored across 7 categories. The minimum passing score is 70%. Correctness below 70% in any critical category (evidence quality, engineering depth) is an automatic fail.

Category	Weight	Description
Evidence Quality	20%	Sources are real, ranked, and traceable. Minimum 70 evidence items across 5 tiers.
Engineering Depth	20%	Component decomposition is complete. Specifications are concrete, not aspirational.
Trade-offs	15%	Every decision shows gains and sacrifices. Net assessment is justified.
Failure Analysis	15%	7 failure modes studied. Each has probability, severity, mitigation.
Manufacturability	10%	Assembly plan is feasible. Yield and throughput are estimated.
Benchmarks	10%	Compared against 5 real production systems. Honest assessment (BETTER/WORSE).
Uncertainty Calibration	10%	Confidence is propagated honestly. Assumptions have falsifiers. Unknowns are visible.
TOTAL	100%	Minimum passing: 70%. Critical categories below 70% = automatic fail.

The Experiment

Compare The Blueprint output vs the PatSnap output side by side. Both receive the same input. Both must produce a complete design for an EV battery pack and thermal management system.

The important question is not:

"Which system is smarter?"

The important question is:

"Which system would an engineering organization trust more?"

Capability	PatSnap Report	The Blueprint
Constraint graph	☐ Not produced	☐ Gate 6: DAG with sensitivity scores
Trade-off engine	☐ Not produced	☐ Gate 7: 7 dimensions, gains/sacrifices
Benchmark score	☐ Not produced	☐ Gate 8: 5 competitors, BETTER/WORSE
Adversarial review	☐ Not produced	☐ Gate 11: 4 reviewers attempt to destroy
Confidence propagation	☐ Not produced	☐ Evidence-weighted, penalty-adjusted
Manufacturing plan	☐ Not produced	☐ Gate 12: assembly steps, yield, throughput
Bill of materials	☐ Not produced	☐ Gate 12: all components, costs, suppliers
Repairability analysis	☐ Not produced	☐ Gate 12: service intervals, spares
Cost model	☐ Not produced	☐ Gate 12: capital, unit cost, break-even
Simulations	☐ Not produced	☐ Gate 10: 6 scenarios, PASS/FAIL
Failure analysis	☐ Discussed	☐ Gate 9: 7 modes, probability, severity
Evidence cited	☐ Thorough	☐ Gate 2: 70+ items, ranked A-I
Uncertainty acknowledged	☐ Explicit	☐ Rule 7: never pretend certainty
Decision tree	☐ Not produced	☐ Alternatives for every component

The PatSnap result is not a failure — it has a different objective. It produces a research report. The Blueprint produces an engineering blueprint. The experiment tests whether the additional structure (constraint graphs, trade-off engines, adversarial review, simulations) produces a more trustworthy output than a well-written research report.

That is the experiment.